Problem Set 9: Covariance, Correlation and Ways of Working

SPEA V706 Math Camp | Module 9 · Thursday, August 13

Part A is covariance and correlation by hand. **Part B** computes the same objects in R and then makes you organize the work three different ways.

Part A. Concepts

A1. Covariance from a joint distribution

Let $X = 1$ if a worker has a college degree and 0 otherwise. Let Y be the number of promotions the worker received in five years. Their joint mass function is:

	$Y = 0$	$Y = 1$	$Y = 2$
$X = 0$	0.20	0.25	0.05
$X = 1$	0.05	0.20	0.25

a. Verify this is a valid joint mass function, then find the two marginal distributions.
b. Compute $E[X]$ and $E[Y]$.
c. Compute $E[XY]$. Only some cells contribute anything; say which and why.
d. Compute $\text{Cov}(X, Y)$ using V3.
e. Compute $\text{Var}(X)$, $\text{Var}(Y)$, and then $\text{Corr}(X, Y)$.
f. Are X and Y independent?

A2. The cross term

Using your answers from A1:

a. Compute $\text{Var}(X + Y)$ with V5.
b. Compute what you would have gotten if you had wrongly assumed the covariance was zero.
c. By how much, and in which direction, does ignoring the cross term mislead you? State the general rule for the direction.

A3. Signs, from stories

For each pair, say whether you expect the covariance to be positive, negative, or near zero, and give one sentence of reasoning.

a. A city's daily ice cream sales and its daily drowning deaths.
b. A used car's mileage and its resale price.
c. A person's shoe size and their score on a statistics exam.
d. Hours of study and exam score, among students who all study between 9 and 11 hours.

Part (d) is doing something different from the others. Say what.

A4. Zero covariance without independence

Let X take the values $-2, -1, 0, 1, 2$, each with probability $1/5$, and let $Y = X^2$.

a. Compute $E[X]$ and $E[Y]$.
b. Compute $E[XY]$. (Note that $XY = X^3$.)
c. Compute $\operatorname{Cov}(X, Y)$.
d. Now show X and Y are *not* independent by comparing $\Pr(Y = 4)$ with $\Pr(Y = 4 \mid X = 2)$.
e. In two sentences: what does covariance fail to see here, and what feature of this distribution makes it fail?

A5. Working the rules

Suppose $\operatorname{Var}(X) = 4$, $\operatorname{Var}(Y) = 9$, and $\operatorname{Cov}(X, Y) = 3$.

a. Compute $\operatorname{Corr}(X, Y)$.
b. Compute $\operatorname{Cov}(2X + 3,\ Y - 1)$ using V4.
c. Compute $\operatorname{Var}(2X - Y)$.
d. Compute the slope you would get from projecting Y on X, that is $\operatorname{Cov}(X, Y)/\operatorname{Var}(X)$. Then compute the slope from projecting X on Y. Why are they different numbers?

Part B. R

```r
pacman::p_load(data.table, ggplot2, fixest, wooldridge)
data(wage1, package = "wooldridge")
dt <- as.data.table(wage1)
```

B1. Covariance and correlation on real data

a. Compute `cov(educ, lwage)` and `cor(educ, lwage)` in `dt`.
b. The covariance is a small number and the correlation is much larger. Does that mean the relationship is weak or strong? Explain what each number is measuring.
c. Compute the correlation between `educ` and `wage` (in levels). Compare it to the correlation with `lwage`. Why are they not the same?

B2. The slope, two ways

a. Compute `cov(educ, lwage) / var(educ)` by hand in `dt`.
b. Run `feols(lwage ~ educ, data = dt)` and read off the coefficient on `educ`.
c. Confirm the two numbers agree. Write one sentence explaining why they must.
d. Interpret the slope. Because the outcome is in logs, roughly what percent wage difference is associated with one more year of schooling?

B3. Recover a correlation you built

```
set.seed(20260813)
n   <- 500
rho <- 0.6

x <- rnorm(n)
e <- rnorm(n)
y <- rho * x + sqrt(1 - rho^2) * e
```

a. Compute `cor(x, y)`. How close is it to `rho`?
b. Repeat with `n <- 50` and again with `n <- 50000`. What happens to how close the estimate lands?
c. What does the pattern in (b) suggest about estimates and sample size?

B4. Correlation is blind to curvature

```
set.seed(20260813)
x2 <- runif(1000, min = -1, max = 1)
y2 <- x2^2
```

a. Compute `cor(x2, y2)`. What do you get?
b. Plot `y2` against `x2` with `ggplot2` and look at it.
c. Reconcile the picture with the number.

B5. The same analysis, three ways

Take the work from B1 and B2 (load the data, compute the two moments, compute the slope, run the regression, save one figure) and organize it three different ways.

a. **The console.** Type the commands one at a time, nothing saved. Then close R without saving the workspace and try to reproduce your slope. What did you lose?
b. **One script.** Put everything in `ps3_analysis.R`. Restart R (Session > Restart R) and run it top to bottom. Does it work from a clean session? If it does not, the usual cause is an object you created in the console and never wrote down.
c. **A pipeline.** Split the work into `code/01_prepare.R` (load the data, build any variables) and `code/02_analyze.R` (moments, regression, figure), with a `main.r` at the top level that sources both in order using `here()`. Restart R and run only `main.r`.

Check yourself. Without notes:

1. What does correlation add that covariance does not?
2. Give an example of two variables that are dependent but uncorrelated.
3. If you double X, what happens to $\mathrm{Cov}(X, Y)$? To $\mathrm{Corr}(X, Y)$?